

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	A	C	A,C	A,B,C,D	A,C,D	A,D	A,B,D
SECTION-II	Q.	1	2	3	4	5					
	A.	1.41 to 1.42	2.50	6.00	4.50	1.25					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	B	B	A,C	B,C,D	A,B,C,D	B,D	A,B,D
SECTION-II	Q.	1	2	3	4	5					
	A.	6.00	-1.00	50.00	4.00	0.28 or 0.29					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	C	B	D	C,D	A,B,C,D	A,C,D	A,C	A,B,C
SECTION-II	Q.	1	2	3	4	5					
	A.	0.50	1.60	2.10	0.08	1.00					

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	D	B	A	A,B,D	B,C	A,B	A,B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,B,D	A,D	A,B,D	A	B	B	C		

PART-2 : CHEMISTRY

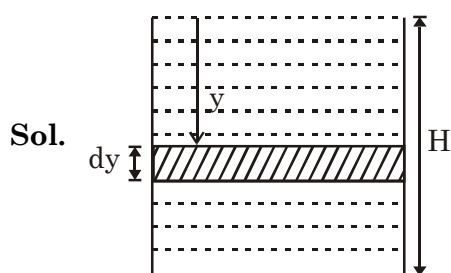
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	C	A	D	A,C	B,C,D	A,B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,B,C	C,D	C	B	D	B	D		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	C	B	D	B,C	A,B,C,D	A,B,D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C,D	A,B,D	B,C,D	B	A	B	B		

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A)



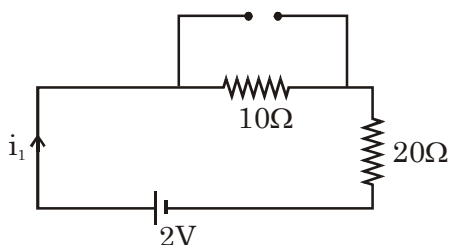
$$dh = \frac{dy}{\mu}$$

$$\int dh = \int_0^H \frac{dy}{\left(1 + \frac{y}{H}\right)}$$

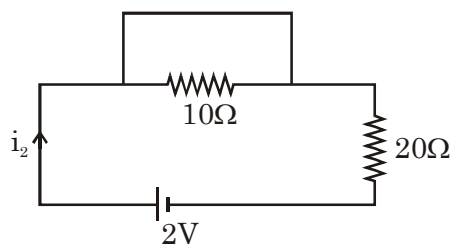
$$h = H \cdot \left[\ln \left(1 + \frac{y}{H} \right) \right]_0^H$$

$$h = H \ln(2)$$

2. Ans. (B)

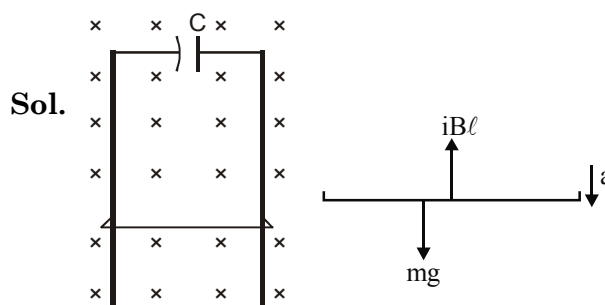
Sol. At $t = 0$ 

$$i_1 = \frac{2}{30} = \frac{1}{15} \text{ A}$$

At $t \rightarrow \infty$ 

$$i_2 = \frac{2}{20} = \frac{1}{10} \text{ A}$$

3. Ans. (D)



$$mg - i\ell B = ma \dots (i)$$

$$\frac{q}{c} = B\ell v \Rightarrow \frac{dq}{dt} = i = B\ell c \frac{dv}{dt} \dots (ii)$$

$$\text{From (i) \& (ii), } a = \frac{mg}{m + B^2 \ell^2 c}$$

4. Ans. (A)

Sol. Flux through

$$S_1(\phi_1) = \frac{\text{Charge enclosed within } S_1}{\epsilon_0} = \frac{q}{\epsilon_0}$$

Since the shell is conducting therefore a charge of magnitude $-q$ will be induced at the inner surface (radius R_1) of the shell.

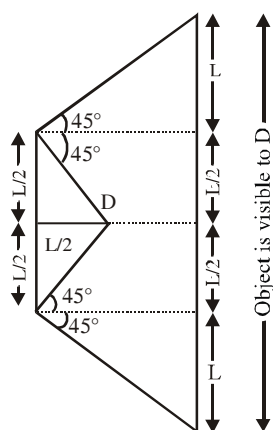
$\therefore$ Flux through

$$S_2(\phi_2) = \frac{\text{Charge enclosed within } S_2}{\epsilon_0} = \frac{q - q}{\epsilon_0} = 0$$

5. Ans. (C)

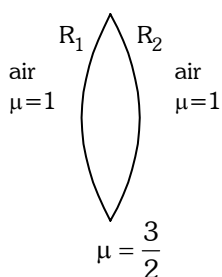
Sol. Required time is given by

$$3L = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{6L}{g}}$$



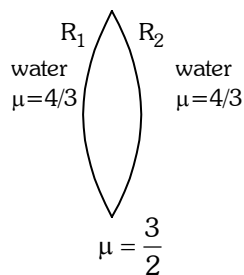
6. Ans. (A,C)

Sol. $\frac{1}{f} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$



$$\frac{1}{f} = \frac{1}{2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\frac{1}{f'} = \left(\frac{\frac{3}{2}}{4/3} - 1 \right) \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$



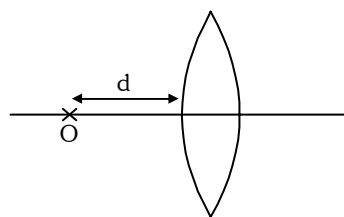
$$\frac{1}{f'} = \left[\frac{9}{8} - 1 \right] \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$$

$$\frac{1}{f'} = \frac{1}{8} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{1}{4f}$$

$$f' = 4f$$

Initial :

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{d}$$



$$\frac{1}{v} = \frac{1}{f} - \frac{1}{d} \Rightarrow v = \frac{fd}{d-f}$$

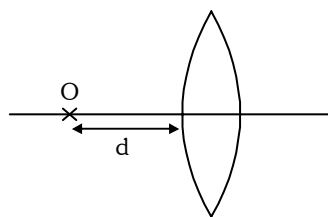
Image is inverted, real and magnified.

$$\therefore v > 0 \Rightarrow d > f \quad \dots(1)$$

$$\text{and } \frac{f}{d-f} > 1$$

$$\begin{aligned} f &> d - f \\ 2f &> d \end{aligned} \quad \dots(2)$$

Final $\frac{1}{4f} = \frac{1}{v'} + \frac{1}{d}$



$$\frac{1}{v'} = \frac{1}{4f} - \frac{1}{d}$$

$$v' = \frac{4fd}{d-4f} < 0 \quad \& \quad \left| \frac{v'}{u'} \right| > 1$$

$\therefore$ image is virtual & magnified.

7. **Ans. (A,B,C,D)**

Sol. (A) $\tau = \vec{M} \times \vec{B} = I\alpha$

$$i\ell^2 B \sin 90^\circ = \frac{2}{3} m \ell^2 \alpha \Rightarrow \alpha = \frac{3Bi}{2m}$$

(B) Initial potential energy

$$= -\vec{M} \cdot \vec{B} = MB \cos 90^\circ = 0$$

Final potential energy

$$= -\vec{M} \cdot \vec{B} = -MB \cos 60^\circ = \frac{-MB}{2} = \frac{-i\ell^2 B}{2}$$

Change in potential energy = Change in rotational kinetic energy

$$\frac{Bi\ell^2}{2} = \frac{1}{2} \times \frac{2}{3} m \ell^2 \times \omega^2$$

$$\omega = \sqrt{\frac{3Bi}{2m}}$$

$$(C) \tau = \vec{M} \times \vec{B} = i\ell^2 B \sin 30^\circ = \frac{Bi\ell^2}{2}$$

$$(D) \tau = MB \sin \theta = I\alpha$$

$$\alpha = \frac{MB \sin \theta}{I}$$

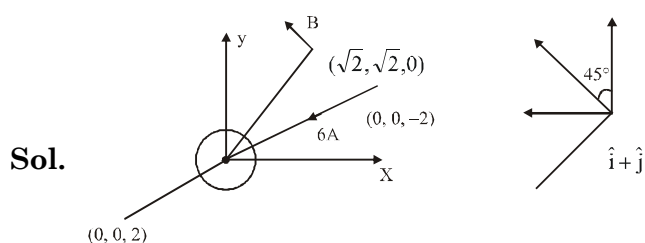
θ increases 90° to 180°

$\Rightarrow \sin \theta$ decreases from 1 to 0

$\therefore \alpha$ decreases

8. **Ans. (A, C, D)**

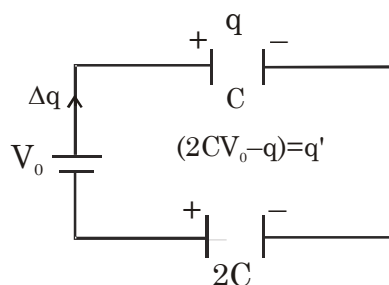
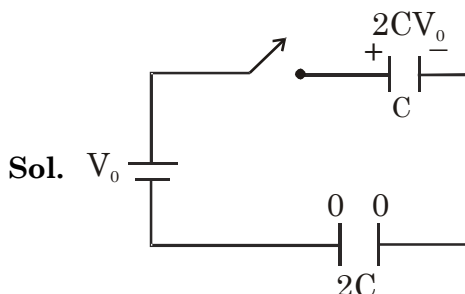
9. **Ans. (A, D)**



$$B = \frac{\mu_0 i}{4\pi r} (\sin 45^\circ + \sin 45^\circ)$$

$$= 3\sqrt{2} \times 10^{-7} \cdot T$$

10. **Ans. (A,B,D)**



$$-V_0 - \left(\frac{2CV_0 - q}{2C} \right) + \frac{q}{C} = 0$$

$$-2V_0 + \frac{q}{2C} + \frac{q}{C} = 0$$

$$\frac{3q}{2C} = 2V_0$$

$$q = \frac{4CV_0}{3}$$

$$q' = 2CV_0 - q = \frac{2CV_0}{3}$$

$$\frac{q}{q'} = 2CV_0 - q = \frac{2CV_0}{3} \dots (2)$$

$$\text{Initial energy } U_i = \frac{1}{2} \cdot C \cdot (2V_0)^2 = 2CV_0^2$$

$$\text{Final energy } U_f = \frac{q^2}{2C} + \frac{(q')^2}{2 \cdot (2C)} = CV_0^2$$

$$\begin{aligned} \text{Work done by battery} \\ = \Delta q \cdot V_0 = (q - 2CV_0)V_0 \end{aligned}$$

$$W_b = \frac{-2CV_0^2}{3}$$

$$\text{Total heat loss} = (U_i - U_f) + W_b = \frac{1}{3}CV_0^2$$

11. Ans. (A,C,D)

$$\text{Sol. Induced current} = \frac{\text{induced emf}}{\text{resistance}}$$

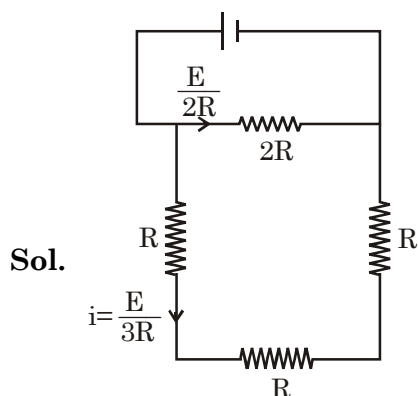
$$= \frac{\varepsilon}{R} = \frac{d\phi}{dtR} = \frac{d(BA)}{dt \times R} = \frac{d(0.4t \times \pi r^2)}{dt \times R}$$

$$= \frac{0.4 \times \pi r^2}{R} = \frac{0.4 \times \pi \times \left(\frac{1}{5}\right)^2}{0.04}$$

$$= \frac{0.4 \times \pi}{0.04} \times \frac{1}{25} = \frac{10\pi}{25} = \frac{2\pi}{5}$$

According to lenz's law field due to ring at center will be in direction opposite to the external magnetic field.

12. Ans. (A,C)



$$Q_{C_2} = C_2 E$$

$$Q_{C_1} = C_1 \times \frac{E}{3}$$

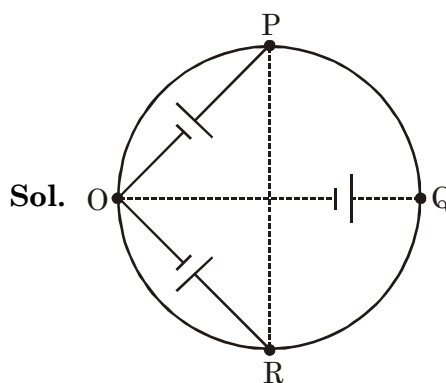
$$Q_{C_3} = C_3 \frac{E}{3}$$

$$Q_{C_4} = C_4 \frac{E}{3}$$

$$C_1 = C_3 = C_4 = 12\mu F$$

$$C_2 = 4\mu F$$

13. Ans. (B, D)



$$V_P - V_0 = \frac{B\omega}{2} \left[(\sqrt{2}R)^2 \right] = B\omega R^2$$

$$V_R - V_0 = \frac{B\omega}{2} \left[(\sqrt{2}R)^2 \right] = B\omega R^2$$

$$V_Q - V_0 = \frac{B\omega}{2} \left[(2R)^2 \right] = 2B\omega R^2$$

SECTION-II

1. Ans. 1.41 to 1.42

$$\text{Sol. } \delta = i + e - A$$

$$30 = 90 - A$$

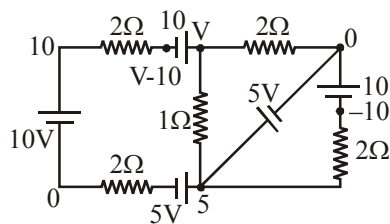
$$A = 60^\circ$$

$$\mu = \frac{\sin\left(\frac{A + \delta m}{2}\right)}{\sin \frac{A}{2}}$$

$$\mu = \frac{\sin 45}{\sin 30} = \sqrt{2}$$

2. Ans. 2.50

Sol.



$$\frac{v-20}{2} + \frac{v}{2} + v - 5 = 0;$$

$$v - 20 + v + 2(v - 5) = 0 = 4v - 20 - 10 = 0$$

$$v = \frac{30}{4}$$

$$I = \frac{v-5}{1} = \frac{15}{2} - 5 = \frac{15-10}{2} = \frac{5}{2}$$

3. Ans. 6.00

Sol. $\vec{\mu}$ (Magnetic moment of loop) when it is

$$\text{lifted by } 45^\circ = i\ell^2 \left(\frac{\hat{j} + \hat{k}}{\sqrt{2}} \right)$$

$\therefore \vec{\tau}$ due to magnetic field

$$= \vec{\mu} \times \vec{B} = \frac{i\ell^2}{\sqrt{2}} \left[(\hat{j} + \hat{k}) \times (3\hat{i} + 4\hat{j}) \right]$$

$\vec{\tau}$ due to mg (about top edge)

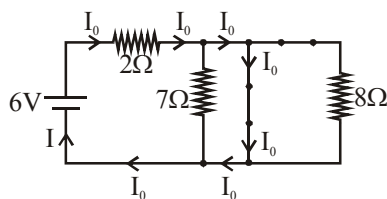
$$= 4mg \frac{\ell}{2} \cos 45^\circ \hat{i}$$

$\therefore$ For equilibrium net torque along X-axis = 0

$$\therefore \frac{4mg\ell}{2\sqrt{2}} = \frac{4i\ell^2}{\sqrt{2}} \Rightarrow i = \frac{mg}{2\ell} = 10A$$

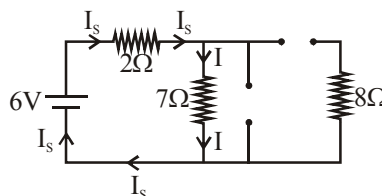
4. Ans. 4.50

Sol. Just after switch is closed, as both capacitors are uncharged, we have current flow as



$$I_0 = \frac{6}{2} = 3A$$

Long time after switch is closed, both capacitor reach their steady state condition. Now the current flow is

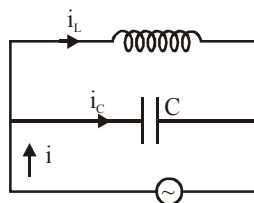


$$I_s = \frac{6}{2+7} = \frac{6}{9} A$$

$$\text{ratio} = \frac{I_0}{I_s} = 4.5$$

5. Ans. 1.25

Sol.



If $V = V_0 \sin \omega t$

$$i_c = \frac{V_0}{X_c} \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$i_L = \frac{V_0}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$i = i_0 + i_L$$

So required ratio = 1.25

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(B)**2. **Ans.(C)**

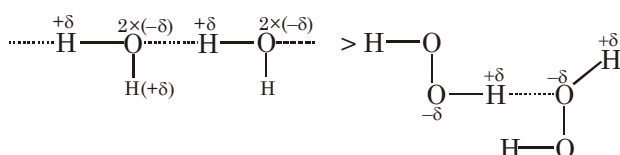
It is example of acid catalytic aldol condensation.

3. **Ans.(D)****Sol.** (A) $\text{NO}^\oplus$, B.O. $\Rightarrow$ 3 $\text{NO}^\ominus$, B.O. $\Rightarrow$ 2

Bond order increases, bond length decreases, $\text{NO}^\oplus < \text{NO}^\ominus$: Bond length

(B) Li_3N exist but K_3N not exist

(C)



Strength of hydrogen bonding the accumulated negative charge on oxygen atom in H_2O is more as compared to that in H_2O_2 .

(D)

$$\text{N}_2^\ominus \Rightarrow (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\pi 2p_x^2 = \pi 2p_y^2) (\sigma 2p_z^1)$$

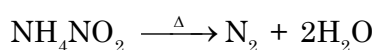
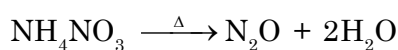
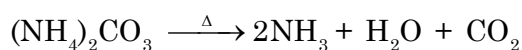
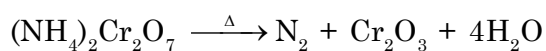
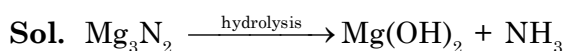
$$\text{B.O.} = \frac{5-0}{2} = 2.5$$

$$\text{N}_2^\ominus \Rightarrow (\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\pi 2p_x^2 = \pi 2p_y^2) (\sigma 2p_z)^2$$

$$(\pi^* 2p_x^1 = \pi^* 2p_y^1)$$

$$\text{B.O.} = \frac{6-1}{2} = 2.5$$

B.O. is same but last electron in $\text{N}_2^\oplus$ is present in BMO so it is more stable than $\text{N}_2^\ominus$.

4. **Ans.(B)**5. **Ans. (B)**

$$\Delta U = q + w$$

$$q = -w (\because \Delta U = 0)$$

$$6000 + (-5500) + (-3000) + 3500 = - [2500 + (-1000 + (-1200) + x)]$$

$$1000 = - [300 + x]$$

$$x = -1300$$

$$\eta = \frac{|W|}{\text{heat abs}} = \frac{1000}{9500} = 0.1052$$

$$= 10.52\%$$

6. **Ans. (A,C)**7. **Ans. (B,C,D)**

$$* t_{1/2} = \frac{[A_0]}{2K} \text{ for zero order.}$$

* For 1st order, $\frac{d[A]}{dt} = k[A]_0 e^{-kt}$ rate decreases with time.

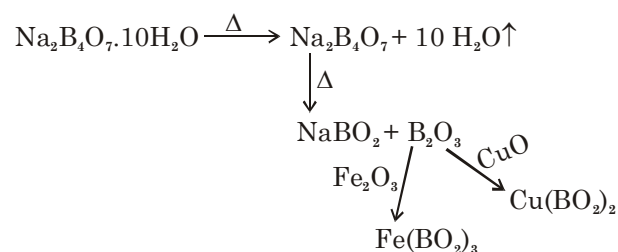
* For 2nd order, $t_{1/2} = \frac{1}{[A]_0 k}$, $t_{1/2}$ decreases with increases in initial concentration.

8. **Ans. (A,B,C,D)**

Sol. Schottky defect is only observed in ionic compound.

9. **Ans.(B,D)**

Sol. $\text{X} = \text{NaCl}$, $\text{Y} = \text{CrO}_2\text{Cl}_2$, $\text{Z} = \text{Na}_2\text{CrO}_4$, Blue solution $\Rightarrow \text{CrO}_5$

10. **Ans.(A,B,D)****Sol.**11. **Ans.(A,B,C,D)**12. **Ans.(A,B,D)**13. **Ans.(B,C,D)**

SECTION-II

1. Ans.(6.00)

2. Ans. -1.00

3. Ans.50.00

$$\Delta T_B = iK_b M$$

$$m = \frac{10 \times 1000}{342(90)} = \frac{1000}{(342)(9)}$$

$$\frac{5}{18} = i \frac{(0.57) \times 1000}{(342)(9)}$$

$$i = \frac{(5)(342)(9)}{(18)(570)} = \frac{(5)(342)}{(1140)} = \frac{171}{114} = 1.5$$

$$\alpha = \frac{i-1}{n-1} = 0.5$$

$$\% \text{ hydrolysis} = 50$$

4. Ans.(4.00)

Sol. $\text{NO}_2^\ominus$, $\text{Br}^\ominus$, $\text{I}^\ominus$, $\text{NO}_3^\ominus$

5. Ans.(0.28 or 0.29)

$$X = 2 (\text{Ag}^+, \text{Hg}_2^{+2})$$

$$Y = 7 (\text{Sn}^{+2}, \text{Hg}^{+2}, \text{Cu}^{+2}, \text{As}^{+3}, \text{Sb}^{+3}, \text{Ag}^+, \text{Hg}_2^{+2})$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

$$\text{Sol. } \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -\hat{i} - 7\hat{j} + 5\hat{k}$$

$$a = -1, b = -7, c = 5$$

$$(-1)(x+1) + (-7)(y+2) + 5(z+1) = 0$$

$$-x - 1 - 7y - 14 + 5z + 5 = 0$$

$$P = 0 \Rightarrow -x - 7y + 5z - 10 = 0$$

2. Ans. (C)

$$\text{Sol. } AA^J = J \Rightarrow \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix} \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix} = I$$

$$\begin{bmatrix} (4b^2 + c^2) & 2b^2 - c^2 & c^2 - 2b^2 \\ 2b^2 - c^2 & a^2 + b^2 + c^2 & a^2 - b^2 - c^2 \\ 2b^2 - c^2 & a^2 + b^2 + c^2 & a^2 - b^2 - c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{compare } 4b^2 + c^2 = 1 ; 2b^2 - c^2 = 0 \text{ \&}$$

$$a^2 + b^2 + c^2 = 1 \Rightarrow b^2 = \frac{1}{6}, c^2 = \frac{1}{3}, a^2 = \frac{1}{2}$$

$$|abc| = \frac{1}{6}$$

3. Ans. (C)

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - 1}{x} - 3x^2 \frac{((1+x^3)^{1/3} - 1)}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{x}{3x} - 3x^2 \frac{\left(1 + \frac{x^3}{3}\right)}{x^3}$$

$$= \frac{1}{3}$$

4. Ans. (B)

$$\text{Sol. } |2(\vec{b} \times \vec{a}) + \vec{a} \times \vec{b}|^2 + 6(\vec{a} \cdot \vec{b})$$

$$|a|^2 |b|^2 \sin^2 \theta + 6|a||b| \cos \theta = 9$$

5. Ans. (D)

$$\text{Sol. } \begin{array}{c} \text{Diagram of a parallelogram with vertices } A(1,2,0) \text{ and } B(-1,1,4). \\ \text{A vector } \vec{AB} \text{ is shown.} \\ \vec{AB} = 2\hat{i} + \hat{j} - 4\hat{k} \end{array}$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -4 \\ 3 & 1 & 5 \end{vmatrix} = 9\hat{i} - \hat{j} + 22\hat{k}$$

$$a = 9, b = -22, c = -1$$

$$9(x-1) - 22(y-2) - 1(z-0) = 0$$

$$9x - 22y - z = -35$$

6. Ans. (C,D)

Sol. $\int_0^1 (f(x) - (f(x^2))^2) dx = \int_0^1 x^2 dx \quad \dots(1)$

for $\int_0^1 f(x) dx$

Let $x = t^2 \Rightarrow dx = 2t dt$

$\int_0^1 f(x) dx = \int_0^1 f(t^2)(2t) dt \quad \dots(2)$

By using (1) & (2)

$\int_0^1 (f^2(x^2) - 2xf(x^2) + x^2) dx = 0$

$\int_0^1 (f(x^2) - x)^2 dx = 0$

$f(x^2) = x \Rightarrow f(x) = \sqrt{x}$

(B) $\int_1^2 \sqrt{x} dx = \frac{x^{3/2}}{3/2} = 2 \frac{(2^{3/2} - 1)}{3}$

(C) $f(x)$ is an increasing & one-one function

7. Ans. (A,B,C,D)

Sol. (A) $\textcircled{CC}, V, A, I, N, E = 6!$

(B) $\underline{A} _ _ \underline{I} _ _ _ = {}^7C_2 \times \frac{5!}{2!}$

$= \frac{7!}{2!5!} \times \frac{5!}{2!}$

(C) $\textcircled{V} \textcircled{A} \textcircled{I} \textcircled{N} \textcircled{E}$

${}^6C_2 \times 5! = 15 \times 5!$

(D) $V _ _ _ _ _ = \frac{6!}{2!}$

8. Ans. (A,C,D)

Sol. $(y+2)^2 = -3\left(x - \frac{4}{3}\right)$

Let $y+2 = Y$ & $x - \frac{4}{3} = X$

$Y^2 = -4\left(\frac{3}{4}\right)(X)$

$X = \frac{3}{4} \Rightarrow x - \frac{4}{3} = \frac{3}{4}$

$x = \frac{3}{4} + \frac{4}{3} = \frac{25}{12}$

$k = 25$

focus is $x - \frac{4}{3} = -\frac{3}{4}$ & $y+2=0$

$x = \frac{4}{3} - \frac{3}{4} = \frac{7}{12}$ & $y = -2$

9. Ans. (A,C)

Sol. $z=0$, in equation of straight line gives

$\frac{x-2}{3} = \frac{y-1}{2} = 1 \Rightarrow x=5, y=3$

$a^2 = 25 - 9 = 16$

$a = \pm 4$

10. Ans. (A,B,C)

Sol. Angle between the lines is

$\cos \theta = \frac{(2\hat{i} - 3\hat{j} + 8\hat{k}) \cdot (\hat{i} - 4\hat{j} + 7\hat{k})}{\sqrt{4+9+64}\sqrt{1+16+49}}$

$= \frac{2+12+56}{\sqrt{77}\sqrt{66}} = \frac{70}{11\sqrt{49}}$

11. Ans. (A,B,D)

Sol. $\ell_1 = \lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2e^{2x} - 2}{2x} = 2$

$\ell_2 = \lim_{x \rightarrow 0} \frac{\sin 3x - 3x}{x^3} = \lim_{x \rightarrow 0} \frac{3(\cos 3x - 1)}{3x^2} = \frac{-3 \sin 3x}{2x}$

$\ell_3 = \lim_{x \rightarrow 0} \frac{\tan x (\cos x - 1)}{x^3} = -\frac{1}{2}$

12. Ans. (B,C)

Sol. $I = \int \frac{x^2 dx}{x^3 (\sqrt{1-x^3})}$

Let $1 - x^3 = t^2$

$3x^2 dx = -2t dt$

$\int \frac{-2t dt}{3t(1-t^2)}$

13. Ans. (B,C)

Sol. $\left(\frac{dy}{dx} - 2^x\right)\left(\frac{dy}{dx} - 2^{-x}\right) = 0$

$$\frac{dy}{dx} = 2^x \text{ or } \frac{dy}{dx} = 2^{-x}$$

Integration

$$y = \frac{2^x}{\ln 2} + C \text{ or } y = \frac{-2^{-x}}{\ln 2} + C$$

SECTION-II

1. Ans. 0.50

Sol. For continuity at $x = 0 \Rightarrow 5 = \alpha$

for continuity at $x = 2 \Rightarrow 14 + \alpha = 2\beta + \gamma$

$$2\beta + \gamma = 19$$

differentiability at $x = 2 \Rightarrow 9 = \beta$

$$\gamma = 1$$

$$\text{LMVT} \Rightarrow \frac{f(3) - f(0)}{3 - 0} = f(c)$$

$$\frac{28 - 5}{3} = \frac{23}{3} = 2c + 5$$

$$2c = \frac{23}{3} - 5$$

$$2c = \frac{8}{3} \Rightarrow c = \frac{4}{3}$$

$$5 + 18 + 3 + 8 = 34$$

2. Ans. 1.60

Sol. Characteristic

$$|A - xI| = 0$$

$$x^3 - 3x^2 - x - 3 = 0$$

$$A^3 - 3A^2 - A - 3I = 0$$

$$(A - I)^3 = 4A + 2I$$

$$(A - I)^3 - 4A = 2I$$

$$|(A - I)^3 - 4A| = 8$$

3. Ans. 2.10

Sol. (1) When one box is empty

$$= {}^3C_1 \left(\frac{3!}{2!1!} \times 2! \right) = 18$$

(2) When two boxes are empty $= {}^3C_2 = 3$

$$\text{Total} = 21$$

4. Ans. 0.08

Sol. $|\vec{a} \times \hat{i}|^2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ 1 & 0 & 0 \end{vmatrix}^2 = a_1^2 + a_2^2$

$$|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2 = 2(a_1^2 + a_2^2 + a_3^2) = 8$$

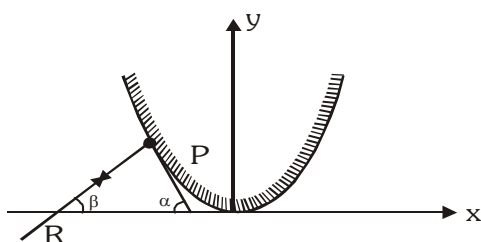
5. Ans. 1.00

Sol. At $x = 0, y = 0$

$$e^{xy} + x e^{xy} (xy' + y) = y + 2 \sin x \cos x$$

put $x = 0, y = 0$, then

$$\frac{dy}{dx} = 1$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A)****Sol.**

Point of intersection of two curves, $y = \frac{x^2}{4}$

and $y = x + 3$ is, $(-2, 1)$

$$\frac{dy}{dx} = \frac{x}{2} \text{ i.e., } \left(\frac{dy}{dx} \right)_{x=-2} = \frac{-2}{2} = -1$$

$\tan \beta = \text{slope of incident ray}$

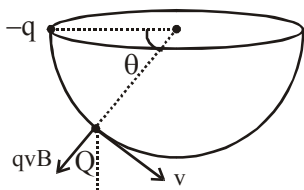
$RP = 1$ or $\beta = 45^\circ$

$$\text{and } \tan \alpha = \left| \left(\frac{dy}{dx} \right)_{x=-2} \right| = 1 \Rightarrow \alpha = 45^\circ$$

From geometry we can show that $\angle i = \angle r = 0^\circ$ i.e., the ray is incident normally.

Hence, the desired unit vector :

$$\mathbf{A} = \frac{1}{\sqrt{2}}(-\hat{i} - \hat{j})$$

2. **Ans. (B)****Sol.**

$$mgR \sin \theta + 0 = \frac{1}{2}mv^2$$

$$N - qvB - mg \sin \theta = \frac{mv^2}{R}$$

$$N = \left(\frac{mv^2}{R} \right) + mg \sin \theta + qvB$$

$$N = 2mg \sin \theta + mg \sin \theta + q\sqrt{2gR \sin \theta} B$$

$$N = 3mg \sin \theta + qB\sqrt{2gR \sin \theta}$$

3. **Ans. (D)**

Sol. Choosing an element of width dx of the triangular conductor at a distance of x from its vertex, area of strip

$$dA = \left(\frac{bx}{h} \right) dx$$

Writing $d\phi = Bda = \{\mu_0 I bx / 2\pi h (a + x)\} dx$
 Integrating between zero and h , we get

$$M = \frac{\phi}{I} = \frac{\mu_0 b}{2\pi h} \{h - a \ln |(a + h)/a|\}$$

$$= 1.2 \times 10^{-8} \text{ H}$$

4. **Ans. (D)**

Sol. Current in circuit :

$$I = \frac{V}{X} = \frac{250}{(75 - 25)} = \frac{250}{50} = 5 \text{ A}$$

$$\therefore V_L = I X_L = 5 \times 25 = 125 \text{ V} \text{ \&}$$

$$V_C = I X_C = 5 \times 75 = 375 \text{ V}$$

voltage on capacitor is more than that of supply voltage because the phase difference between V_L and V_C is 180° (i.e. out of phase)

5. **Ans. (B)**

Sol. Current through the inductor before closing the switch = 1 A

Current through the inductor after closing the switch (in steady state)

$$I = \frac{20}{5} = 4 \text{ A}$$

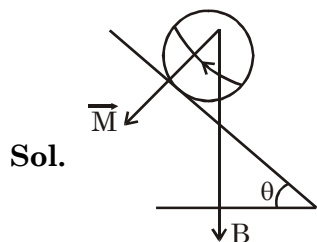
$$\therefore \Delta\phi = LI = 1.5 \text{ wb}$$

6. **Ans. (A)**

Sol. $\frac{\mu_0 i}{4\pi R}(-\hat{j}) - \frac{\mu_0 i}{4R}(-\hat{k})$

7. **Ans. (A, B, D)**

8. **Ans. (B, C)**

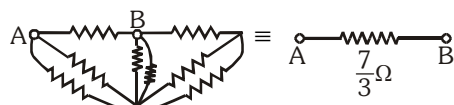


$$MB \sin\theta = mg \sin\theta \times R$$

$$B = \frac{mg}{\pi i R}$$

9. **Ans. (A, B)**

Sol. earranged circuit between A & B is :



(due to symmetry)

Total resistance of circuit

$$= \frac{7}{3} + \frac{2}{3} = 3 \Omega . i = \frac{9}{3} = 3 \text{ A};$$

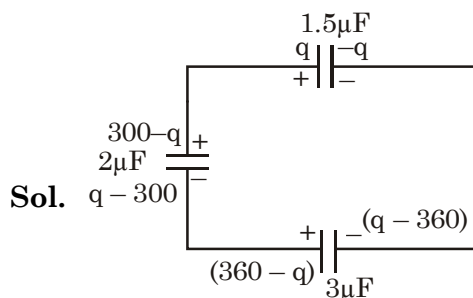
Heat produced in cell

$$= I^2 r = (3)^2 \times \left(\frac{2}{3}\right) = 6 \text{ W}$$

Current in resistance connected directly

$$\text{between A \& B} = \frac{7}{15} \times 3 = \frac{7}{5} = 1.4 \text{ A}$$

10. **Ans. (A, B)**



by applying K.V.L. in a loop

$$-\frac{2q}{3} + \frac{360-q}{3} + \frac{300-q}{2} = 0$$

$$q = 180$$

Potential drop of $2\mu\text{F}$ capacitor is

$$\Rightarrow \frac{120}{2} = 60 \text{ V}$$

11. **Ans. (A, B, C, D)**

Sol. before switch is open

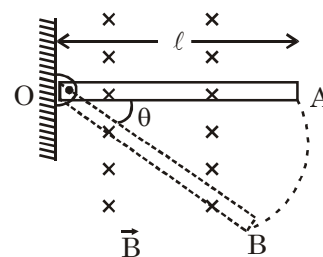
$$v_{3\mu\text{F}} = 9$$

$$v_{2\mu\text{F}} = 4.5$$

$$\phi_{3\mu\text{F}} = 27 \mu\text{C}$$

$$\phi_{2\mu\text{F}} = 9 \mu\text{C}$$

12. **Ans. (A, B, D)**



Sol.

WET from A to B

$$\omega_{\text{mg}} = \Delta \text{ K.E.}$$

$$mg \frac{L}{2} \sin\theta = \frac{1}{2} I \omega^2$$

$$mg \frac{L}{2} \sin\theta = \frac{1}{2} \times \frac{mL^2}{3} \omega^2$$

$$\frac{3g}{L} \sin\theta = \omega^2$$

Induced emf produced in rod is

$$\varepsilon = \frac{1}{2} (B\omega L^2) = \frac{1}{2} B \times \sqrt{\frac{3g \sin\theta}{L}} \times L^2$$

$$\varepsilon = \frac{1}{2} \times B \times \sqrt{\frac{3g \sin\theta \times L^4}{L}}$$

$$\varepsilon = \frac{1}{2} B \sqrt{3g \sin\theta} L^3$$

$$\varepsilon \propto L^{3/2}, \varepsilon \propto \sqrt{\sin\theta}$$

13. Ans. (A,D)

Sol. By using principle of reversibility just interchange position of object and image.

$$\frac{3/2 - 1}{v_1 - R} = \frac{3/2 - 1}{R}$$

{Refraction from Ist surface}

$$\frac{3}{2v_1} = \frac{1}{2R} \frac{-1}{R} = \frac{-1}{2R}$$

$$v_1 = -3R$$

Refraction from IInd surface

$$u_2 = -5R$$

$$\frac{1}{x} - \frac{3/2}{-3R} = \frac{1 - 3/2}{-R}$$

$$\frac{1}{x} + \frac{3}{10R} = \frac{1}{2R}$$

$$\frac{1}{x} = \frac{1}{2R} - \frac{3}{10R} = \frac{5R - 3}{10R} = \frac{1}{5R}$$

$$x = 5R$$

Distance of object from centre of sphere is

$$5R + R = 6R$$

14. Ans. (A,B,D)

Sol. $i_{rms} = 5A$

$$\text{Reading of voltmeter} = i_{rms} \sqrt{R^2 + X_C^2}$$

$$\text{From this } X_C = 20\Omega \Rightarrow \omega = 10^3 \text{ rad/s}$$

$$\Rightarrow f = \frac{1000}{2\pi}$$

$$\therefore X_L = \omega L = 20\Omega$$

therefore the circuit is in resonance

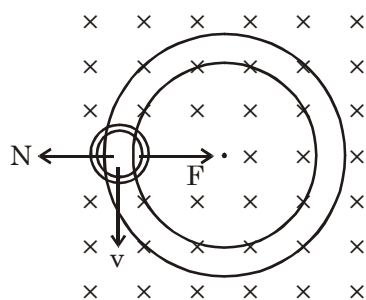
$$E_{rms} = i_{rms} R = 50 \text{ V also power factor} = 1$$

$$\text{av. Power} = E_{rms} i_{rms} \cos \phi = 250 \text{ W}$$

15. Ans. (A)

16. Ans. (B)

Sol. (15 & 16)



$$F - N = \frac{mv^2}{R}$$

$$N = F - \frac{mv^2}{R}$$

$$N = qvB - \frac{mv^2}{R}$$

Induced electric field

$$E = \frac{R}{2} \frac{dB}{dt} = \frac{R}{2} \alpha$$

$$v = \left(\frac{qE}{m} \right) t = \frac{q}{m} \frac{R}{2} \alpha t$$

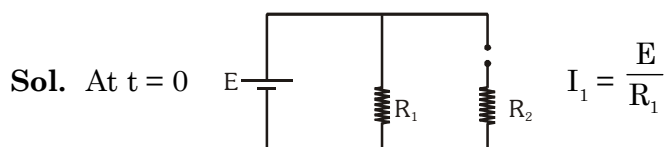
$$N = qvB - \frac{mv^2}{R}$$

$$= qB_0(1 + \alpha t) \left(\frac{qR}{2m} \alpha t \right) - \frac{m}{R} \frac{q^2}{m^2} \frac{R^2 \alpha^2 t^2}{4}$$

$$= \frac{q^2 R \alpha t}{2m} \left[B_0 (1 + \alpha t) - \frac{\alpha t}{2} \right]$$

$$= \frac{q^2 R \alpha t}{2m} \left[B_0 + \frac{\alpha t}{2} \right] = \frac{q^2 R \alpha t}{4m} [2B_0 + \alpha t]$$

17. Ans. (B)



$$= \frac{25}{5} = 5A$$

18. Ans. (C)

$$\text{Sol. } I_2 = \frac{E}{R_2} = \frac{25}{10} = 2.5 \text{ A}$$

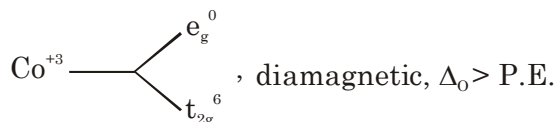
PART-2 : CHEMISTRY

SOLUTION

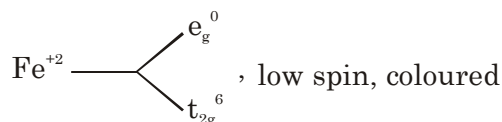
SECTION-I

1. **Ans.(C)**

Sol. (A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$



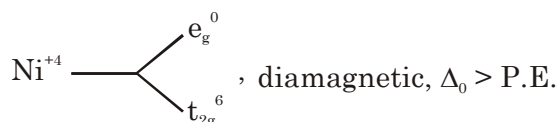
(B) $\text{K}_4[\text{Fe}(\text{CN})_6]$



(C) $[\text{Co}(\text{H}_2\text{O})_6]\text{SO}_4$



(D) $[\text{NiF}_6]^{-2}$



2. **Ans.(B)**

Sol. $\text{A} \Rightarrow \text{NaNO}_2 / \text{NaNO}_3$

$\text{B} \Rightarrow \text{NH}_3, \text{C} \Rightarrow \text{NH}_4\text{Cl}$

$\text{D} \Rightarrow [\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

3. **Ans.(C)**

$$\lambda \times t = \ln \frac{A_0}{A_t}$$

$$\left(\frac{\ln 2}{14.2} \right) \times t = \ln \frac{1000}{1}$$

$$t = \frac{\ln 1000}{\ln 2} \times 14.2$$

$$= \frac{3}{0.3} \times 14.2 = 142 \text{ days}$$

4. **Ans.(C)**

$$\frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right] = R \left[\frac{n^2 - 4}{4n^2} \right]$$

$$\lambda = \frac{4}{R} \times \left(\frac{n^2}{n^2 - 4} \right) = K \times \left(\frac{n^2}{n^2 - 4} \right)$$

On comparing, we get

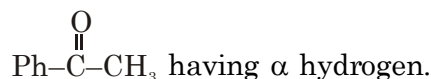
$$K = \frac{4}{R}$$

5. **Ans.(A)**

Perkin reaction

6. **Ans.(D)**

Carbonyl compound having αH can give aldol condensation reaction. Only acetophenone



7 **Ans.(A,C)**

8. **Ans.(B,C,D)**

9. **Ans.(A,B,C,D)**

10. **Ans.(A, B, D)**

11. **Ans.(A, B)**

$$\left(\frac{\partial H}{\partial T} \right)_P = C_P$$

$$\left(\frac{\partial U}{\partial T} \right)_V = C_V$$

$$C_P > C_V$$

$$\left(\frac{\partial H}{\partial P} \right)_T = 0$$

$$\left(\frac{\partial U}{\partial P} \right)_T = 0$$

12. Ans.(A,B,C)

$$(A) \frac{D}{d} = \alpha + 1$$

$$y = x - 1$$

$$(B) K_C = \frac{[NO_2]^2}{[N_2O_4]}$$

$$[N_2O_4] \propto [NO_2]^2$$

$$y \propto x^2$$

$$(C) [NO_2] \propto \sqrt{[N_2O_4]}$$

$$y \propto \sqrt{x}$$

$$(D) \ln K_P = \left(\frac{-\Delta H}{R} \right) \frac{1}{T} + \frac{\Delta S}{R}$$

$$\Delta H > 0, m < 0$$

13. Ans.(C,D)

Sol. Layer test is given by $Br^\ominus$ & $I^\ominus$

14. Ans.(C)

Sol. SO_3^{2-} is act as reducing agent

15. Ans.(B)

16. Ans.(D)



17. Ans.(B)

$$[CH_3COOH] = \frac{6}{60} = 0.1M$$

$$pH = \frac{1}{2}(5 - \log 0.1) = 3$$

18. Ans.(D)

$$pH = 7 + \frac{1}{2}(pK_a + \log C)$$

$$= 7 + \frac{1}{2}[5 + \log(0.05)]$$

$$= 7 + \frac{1}{2}[5 + \log 5 - 2]$$

$$= 7 + \frac{1}{2}[3.7] = 8.85$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (D)

Sol. $\lim_{n \rightarrow \infty} \frac{(n^2 + 1)^{1/2} + n^{1/2}}{(n^3 + n)^{1/4} - n^{1/4}}$

$$\lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n^2}\right)^{1/2} + \frac{1}{\sqrt{n}}}{\left(\frac{1}{n} + \frac{1}{n^3}\right)^{1/4} - \frac{1}{n^{3/4}}} \Rightarrow \text{Not exist}$$

2. Ans. (C)

Sol. $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{36^x - 9^x - 4^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$

$$\lim_{x \rightarrow 0} \frac{(9^x - 1)(4^x - 1)}{(1 - \cos x)} (\sqrt{2} + \sqrt{1 + \cos x})$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{9^x - 1}{x}\right) \left(\frac{4^x - 1}{x}\right) (\sqrt{2} + \sqrt{1 + \cos x})}{\left(\frac{1 - \cos x}{x^2}\right)}$$

$$2(\ln 9 \cdot \ln 4 \cdot 2\sqrt{2})$$

$$\Rightarrow 16\sqrt{2} \ln 2 \ln 3$$

3. Ans. (C)

Sol. $f(x) = \sin(4\pi[x])$

Since $\sin n\pi = 0$, where 'n' is any integer

$$\therefore f(x) = 0 \quad \forall x \in \mathbb{R}$$

4. Ans. (C)

Sol. $y = \sin^2 x + \cos 2(\alpha + \beta) +$

$$(\cos(\alpha - \beta) - \cos(\alpha + \beta))\cos(\alpha + \beta)$$

$$y = \sin^2 \alpha + \cos^2(\alpha + \beta) + (\cos^2 \alpha - \sin^2 \beta) - \cos^2(\alpha + \beta)$$

$$y = \cos^2 \beta$$

$$\frac{dy}{d\alpha} = 0 \Rightarrow \frac{d^3y}{d\alpha^3} = 0$$

5. **Ans. (B)**

Sol. Curve $y = e^{-2|x|}$ cuts the line $x = \frac{1}{2}$ at

$$P\left(\frac{1}{2}, \frac{1}{e}\right)$$

$$\therefore \frac{dy}{dx} = e^{-2x} \cdot (-2)$$

$$\therefore \text{Slope of normal at point } P = \frac{e}{2}$$

equation of normal at P :

$$y - \frac{1}{e} = \frac{e}{2} \left(x - \frac{1}{2} \right)$$

$$4ey - 4 = 2e^2x - e^2$$

$$2e^2x - 4ey = e^2 - 4$$

6. **Ans. (D)**

Sol. $f(x) = -x^3 + 4ax^2 + 2x - 5$

$$f'(x) = -3x^2 + 8ax + 2$$

since $f(x)$ is decreasing for all 'x'

$$\therefore f'(x) < 0 \quad \forall x \in \mathbb{R}$$

$$\therefore -3x^2 + 8ax + 2 < 0 \quad \forall x \in \mathbb{R}$$

$$\therefore \text{Discriminant} < 0$$

$$64a^2 + 24 < 0 \Rightarrow a \in \mathbb{Q}$$

7. **Ans. (B,C)**

$$\text{Sol. } \int \frac{dx}{(1+\sqrt{x})^8}$$

$$\text{Let } \sqrt{x} = t \Rightarrow x = t^2 \Rightarrow dx = 2t dt$$

$$\int \frac{2t dt}{(1+t)^8}$$

$$2 \left[\int \frac{dt}{(1+t)^7} - \int \frac{dt}{(1+t)^8} \right]$$

$$2 \left[\frac{-1}{6(1+t)^6} + \frac{1}{7(1+t)^7} \right]$$

8. **Ans. (A,B,C,D)**

Sol. $B = B^2 \quad \dots(1)$

$$\text{since } A = I - B \Rightarrow AB = B - B^2 = 0$$

$$\therefore A^2 = A(I - B) = A - AB = A$$

$\Rightarrow A$ is also idempotent matrix

$$BA = B - B^2 = 0$$

9. **Ans. (A,B,D)**

$$\text{Sol. } x \cos x \frac{dy}{dx} + y(x \sin x + \cos x) = 1$$

$$\frac{dy}{dx} + \frac{(x \sin x + \cos x)}{x \cos x} y = \frac{1}{x \cos x}$$

$$\text{I.F.} = e^{\int \frac{(x \sin x + \cos x)}{x \cos x} dx}$$

$$= e^{\int \left(\tan x + \frac{1}{x} \right) dx}$$

$$= e^{\ell \sec x + \ell \ln x}$$

$$= x \sec x$$

$$\therefore y \cdot x \sec x = \int \sec^2 x dx + C$$

$$xy \sec x = \tan x + C$$

$$xy = \sin x + \cos x$$

10. **Ans. (A)**

$$\text{Sol. } \vec{u} \times \vec{v} = (\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})$$

$$= \vec{a} \times \vec{b} - \vec{b} \times \vec{a} = 2\vec{a} \times \vec{b}$$

$$= 2\sqrt{|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2}$$

$$= 2\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$$

11. **Ans. (C)**

Sol. vertex $\equiv (0, 2)$ and extremities of L.R. are $(-6, 4)$ & $(6, 4)$

$$\therefore \text{length of L.R.} = 4a = 12$$

$$\text{equation of parabola} \Rightarrow x^2 = 12(y - 2)$$

12. **Ans. (C,D)**

Sol. $f''(x) > 0 \Rightarrow f'(x)$ is increasing

$$f'(C_1) = \frac{f(1) - f(0)}{1}, C_1 \in (0, 1) \text{ and}$$

$$f'(C_2) = \frac{f(2) - f(1)}{2-1}, C_2 \in (1, 2)$$

$$\Rightarrow f'(C_1) < f'(C_2)$$

$$\Rightarrow f(1) - f(0) < f(2) - f(1)$$

$$2f(1) < f(0) + f(2)$$

Similarly applying LMVT between $\left[0, \frac{2}{3}\right]$

and $\left[\frac{2}{3}, 2\right]$

$$\frac{f(2) - f(2/3)}{2 - \frac{2}{3}} > \frac{f(2/3) - f(0)}{\frac{2}{3} - 0}$$

$$f(2) - f(2/3) > 2f(2/3) - 2f(0)$$

$$f(2) + 2f(0) > 3f(2/3)$$

13. Ans. (A,B,D)

Sol. $\text{adj}(AB) = \text{adj}(B) \cdot \text{adj}(A)$

$$= |B| |A| \frac{\text{adj} B}{|B|} \cdot \frac{\text{adj} A}{|A|}$$

$$= |B| |A| B^{-1} \cdot A^{-1} = |B| |A| (AB)^{-1}$$

14. Ans. (B,C,D)

$$\begin{aligned} \text{Sol. } I &= \int_0^1 \frac{\sqrt{1+\sqrt{x}}}{\sqrt{1-\sqrt{x}}} dx = \int_0^1 \frac{1+\sqrt{x}}{\sqrt{1-x}} dx \\ &= \int_0^1 \frac{1}{\sqrt{1-x}} dx + \int_0^1 \frac{\sqrt{x}}{\sqrt{1-x}} dx \end{aligned}$$

$$= \int_0^1 \frac{dx}{\sqrt{x}} + \int_0^1 \frac{\sqrt{1-x}}{\sqrt{x}} dx$$

$$x = t^2 \Rightarrow dx = 2t dt$$

$$= \left[2\sqrt{x} \right]_0^1 + \int_0^1 \frac{\sqrt{1-t^2}}{t} \cdot 2t dt$$

$$= 2 + 2 \int_0^1 \sqrt{1-t^2} dt$$

$$= 2 + 2 \left[\frac{1}{2} t \sqrt{1-t^2} + \frac{1}{2} \sin^{-1} t \right]_0^1$$

$$= 2 + \left[\frac{\pi}{2} - 0 \right]$$

$$= 2 + \frac{\pi}{2}$$

$$\therefore J = \int_0^1 \frac{\sqrt{1-\sqrt{x}}}{\sqrt{1+\sqrt{x}}} dx = \int_0^1 \frac{1-\sqrt{x}}{\sqrt{1-x}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{1-x}} dx - \int_0^1 \frac{\sqrt{1-x}}{\sqrt{x}} dx$$

$$= 2 - \frac{\pi}{2}$$

$$I - J = \pi$$

Paragraph for Question 15 & 16

15. Ans. (B)

Sol. Let number of sides of polygon = n

$$\therefore {}^nC_2 - n = 35$$

$$n = 10$$

$\therefore$ Number of triangles formed

$$= {}^nC_3 = {}^{10}C_3 = 120$$

16. Ans. (A)

Sol. Number of points of intersection of diagonals

which lies inside the polygon = nC_4

$$= {}^{10}C_4 = 210$$

Paragraph for Question 17 & 18

17. Ans. (B)

Sol. $\vec{A} \cdot \vec{B} = 0, |\vec{A}| = |\vec{B}| = 1$

$$\vec{P} \times \vec{B} = \vec{A} \cdot \vec{P}$$

$$\Rightarrow \vec{B} \cdot (\vec{P} \times \vec{B}) = \vec{B} \cdot \vec{A} - \vec{P} \cdot \vec{B}$$

$$0 = 0 - \vec{P} \cdot \vec{B}$$

$$\vec{P} \cdot \vec{B} = 0$$

$$\therefore (\vec{P} \times \vec{B}) \times \vec{B} = (\vec{P} \cdot \vec{B}) \vec{B} - (\vec{B} \cdot \vec{B}) \vec{P} = -\vec{P}$$

18. Ans. (B)

Sol. $(\vec{P} \times \vec{B}) \times \vec{B} = \vec{A} \times \vec{B} - \vec{P} \times \vec{B}$

$$\Rightarrow -\vec{P} = \vec{A} \times \vec{B} - \vec{A} + \vec{P}$$

$$\vec{P} = \frac{1}{2} (\vec{A} - \vec{A} \times \vec{B})$$